T.4: ApportionRegions: Redistricting as the Geographic Completion of Huntington-Hill

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Abstract

The U.S. Constitution allocates congressional seats to states via the Huntington-Hill apportionment method (Article I, §2), but treats the drawing of district boundaries as a separate "Manner of election" question (Article I, §4). We propose collapsing this distinction. Given that a state receives n seats, we construct its congressional map by factoring n into primes and applying those factors hierarchically as k-way geographic partitions of the state's census-tract graph. The resulting *ApportionRegions* (AR) algorithm has two inputs: the prime factorization of the seat count (arithmetic) and the minimum-cut partition of the geographic graph (TIGER adjacency). All prime-p splits of a given state are computed once and reused across every seat count sharing that prime factor. When n changes after reapportionment, only the levels of the factorization tree affected by the new prime structure require recomputation. Under a fixed implementation, seed derivation rule, and deterministic tie-breaking policy, the map is determined by census data and number theory rather than partisan inputs or manual boundary choices. We argue that AR can be understood as an Article I §2 apportionment extension, evaluate the algorithm on all 50 states, and characterize the striking discontinuities that occur when consecutive seat counts have disjoint prime factorizations (e.g., California's $51 = 3 \times 17$ and $52 = 2^2 \times 13$ maps share no structural elements).

1 Introduction

The United States Constitution establishes congressional representation through two distinct clauses. Article I, §2 mandates that seats be *apportioned among the several States* according to population, a requirement implemented since 1941 by the Huntington-Hill method [Balinski and Young, 2001]. Article I, §4 provides that "the Times, Places and Manner of holding Elections ... shall be prescribed in each State by the Legislature thereof," but that "Congress may at any time by make or alter such Regulations." In practice, these clauses are treated as governing separate questions: §2 determines *how many* seats each state receives; §4 (via state or congressional action) determines *where* the district boundaries lie.

We argue this separation is artificial and consequential. The seat count n assigned to a state is not merely a quota — it encodes a geometric structure. Specifically, the prime factorization of n determines a canonical largest-prime-first split prescription for any geographic region: split first by the largest prime factor, then recurse on each child workload, using a binary floor/ceiling fallback when the remaining workload is itself a large prime. When applied to a state's census-tract graph using minimum-cut graph partitioning, this hierarchy produces a congressional map with two inputs only:

1. The prime factorization of n — pure arithmetic, determined by the apportionment process.

2. The census-tract adjacency graph with TIGER boundary weights — pure geography, derived from federal public records.

No political data enters. No human boundary choices are made once the implementation, seed derivation rule, and tie-breaking policy are fixed. The resulting *ApportionRegions* (AR) algorithm is a direct geographic extension of Huntington-Hill: the same constitutional process that decides how many seats a state receives also decides, without further human input, where those seats are located.

The reuse property. The most practically significant feature of AR is that geographic splits are computed once and reused. The canonical largest-prime split is computed once and reused across seat counts that share that same top-level factor. For example, seat counts $34 = 2 \times 17$ and $51 = 3 \times 17$ both begin with the same canonical 17-way partition before their child workloads diverge. When a state gains or loses a seat after reapportionment, only the levels of the factorization tree affected by the changed prime structure require recomputation. Districts have *ancestors* in the tree — a district in the n-seat map can be traced to the leaf of the factorization tree that contains its census tracts, and that leaf's ancestry is preserved across all seat counts sharing the same prime structure at that level.

The discontinuity problem. AR also makes visible a previously unrecognized structural phenomenon: consecutive seat counts can have disjoint prime factorizations, causing complete structural breaks in the map. California currently has 52 seats ($52 = 2^2 \times 13$); the preceding apportionment gave it 53 seats (53 is prime). Going from 52 to 53 requires discarding the 13-way top-level structure and replacing it with the large-prime binary fallback $53 \to (26, 27)$. Going from 51 ($= 3 \times 17$) to 52 ($= 4 \times 13$) replaces a 17-way top level with a 13-way top level. These discontinuities are *mathematically forced*, not arbitrary: they reflect the prime structure of the integers, not political choices. We characterize when such breaks occur and propose a continuity-preserving variant (AR-smooth) that minimises structural change across reapportionments.

Contributions.

1. **Algorithm**: The AR algorithm — a deterministic, prime-factorization-guided redistricting procedure with formal reuse semantics and a two-input specification (prime factorization + census geography).

2. **Constitutional argument**: An argument that AR maps can be understood as extending Article I §2 (apportionment) into geography, rather than treating §4 (Manner) as the only source of districting authority.
3. **Reuse theorem**: A proof that prime-p splits are canonical and reusable, and that the factorization tree is unique given the minimum-cut at each level.
4. **Empirical evaluation**: Full AR maps for all 50 states at their 2020 apportionment seat counts, comparison to minimum-edge-cut (MEC) maps from Della-Libera [2026], and characterisation of the 51→52 California discontinuity.
5. **AR-smooth**: A reapportionment-stable variant that preserves as much of the existing factorization tree as possible when n changes by ± 1.

2 Related Work

Apportionment. The Huntington-Hill method [Huntington, 1921, Hill, 1911] allocates congressional seats using a geometric-mean priority rule: a state with n seats receives the next seat before any state with fewer if its priority $P/\sqrt{n(n+1)}$ (where P is population) exceeds all competitors. Balinski and Young [2001] provide the canonical treatment of apportionment methods and their fairness properties. Crucially, no prior work connects the arithmetic structure of the resulting seat counts to the geometry of the districts those seats represent.

Graph partitioning. The k-way graph partitioning problem — partition the vertices of a weighted graph into k balanced parts minimising edge cut — is NP-hard [Garey and Johnson, 1979]. The METIS multilevel scheme [Karypis and Kumar, 1998] provides practical heuristics for both recursive bisection ($k = 2^d$) and direct k-way partitioning. Our prior work [Della-Libera, 2026] established minimum-edge-cut (MEC) redistricting using recursive bisection and characterised its seed sensitivity. AR generalises MEC by using *prime*-way splits, not only binary ones.

Hierarchical redistricting. Existing graph-partitioning redistricting methods use recursive structure as an implementation strategy: recursive bisection repeatedly splits the current region, and k-way partitioning splits a region directly into k balanced parts. AR makes a different claim. It treats the branching factor and recursion depth as consequences of the apportioned seat count's prime factorization, not as tunable engineering choices.

Constitutional basis. U.S. Supreme Court [2019] (Rucho v. Common Cause) held that partisan gerrymandering claims are non-justiciable in federal court but did not foreclose congressional regulation of the manner of elections. U.S. Supreme Court [2023] (Moore v. Harper) rejected the independent state legislature doctrine, confirming that both state and federal law constrain redistricting. AR's constitutional argument anchors redistricting in Article I §2 (apportionment) rather than §4 (Manner), a distinction unexplored in the legal literature.

Algorithmic redistricting. Duchin and Walch [2019] propose Markov-chain sampling of the redistricting plan space. Herschlag et al. [2020] use sampling to quantify gerrymandering in North Carolina. Chikina et al. [2017] develop statistical significance tests for partisan plans. AR differs from all of these by not sampling a plan space at all: it produces a single canonical plan from arithmetic and geography under a disclosed implementation, seed rule, and tie-breaking policy.

3 Methodology

3.1 Notation

Let $G = (V, E, w)$ be the census-tract adjacency graph for a state, where V is the set of census tracts, E is the set of shared-boundary edges, and $w : E \to \mathbb{R}_{>0}$ assigns each edge the TIGER boundary length between adjacent tracts. Let $\mathrm{pop} : V \to \mathbb{Z}_{>0}$ be the 2020 census population of each tract. Let n be the state's apportioned congressional seat count.

3.2 The Split Prescription

At each level of the AR tree, a region with k target districts is split according to the following rule with threshold $\kappa = 3$:

Definition 1 (Split prescription)**.** *For $k \geq 1$, split(k) is:*

1. ***Direct*** *($k \leq \kappa$): split into k equal parts (sub-regions each with target 1).*
2. ***Uniform-composite*** *($k > \kappa$, k composite): let p be the* largest *prime factor of k. Split into p equal parts, each with target k/p.*
3. ***Binary fallback*** *($k > \kappa$, k prime): split into 2 parts with population targets $\lfloor k/2 \rfloor$ and $\lceil k/2 \rceil$.*

Examples: $\mathrm{split}(8) = (2, \mathrm{sub} = 4)$; $\mathrm{split}(9) = (3, \mathrm{sub} = 3)$; $\mathrm{split}(17) = (2, \mathrm{sub} = (8, 9))$; $\mathrm{split}(34) = (17, \mathrm{sub} = 2)$; $\mathrm{split}(51) = (17, \mathrm{sub} = 3)$.

Why largest prime first? The reuse property requires that two seat counts n and n' sharing a common factor p produce the same top-level p-way partition. With largest-prime-first, $n = 34 = 2 \times 17$ and $n' = 51 = 3 \times 17$ both prescribe a 17-way primary split, sharing the same 17-district top-level partition of any given state. Smallest-prime-first would instead bisect the state first (different halves for $n = 34$ vs. thirds for $n' = 51$), destroying reuse.

Why binary fallback for large primes? A direct k-way METIS cut for large prime k (e.g. $k = 17$) is computationally feasible but produces no reuse with neighbouring seat counts and is harder to balance than iterated bisections. The binary fallback $17 \to (8, 9)$ naturally decomposes into the highly-reused sub-trees $k = 8 = 2^3$ (three binary levels) and $k = 9 = 3^2$ (two ternary levels).

3.3 The AR Algorithm

Definition 2 (ApportionRegions)**.** *Given graph G and seat count n, the AR map $M_n(G)$ is constructed by the recursive procedure $AR(G, n)$:*

1. *If $n = 1$: return $\{G\}$ (one district = whole region).*
2. *Compute $split(n) = (p, k_1, \ldots, k_p)$ where p is the number of parts and k_i is the target district count for part i (Definition 1).*
3. *Partition G into p sub-regions $R_1, \ldots, R_p$ with target population fractions k_i/n each, minimising edge-cut weight.*
4. *For each i: recursively compute $AR(R_i, k_i)$.*
5. *Return the union of all recursive results.*

The p-way partition in step 3 is computed by METIS 5.2 (version with `part_kway` for $p > 2$, `part_recursive` for $p = 2$) using population vertex weights and TIGER boundary-length edge weights. For the uniform case ($k_1 = \cdots = k_p = n/p$) METIS uses equal target fractions; for the binary fallback case ($k_1 = \lfloor n/2 \rfloor$, $k_2 = \lceil n/2 \rceil$) it uses proportional target fractions k_1/n and k_2/n.

In the atlas framing, each AR node exposes three inspectable candidate objects: the arithmetic split prescription, the METIS partition for that prescription, and the recursive child workload list. The decision rule is fixed by the seat-count factorization before geography is consulted; METIS then constructs the geographic realisation of that arithmetic choice. The consequence is a tree of reusable subregions, so the transcript should record the factor used at every node, the child target counts, the seed, and the partition hash. That evidence separates AR's structural claim from any later plan-quality or partisan-outcome claim.

Determinism assumption. METIS 5.2's multilevel k-way partitioning is deterministic given a fixed random seed and fixed input graph (adjacency structure, vertex weights, edge weights). We rely on this property throughout: when we say two seat counts "share the same partition," we mean the assignment of census tracts to districts is byte-identical, conditional on the same METIS seed at that level. Our implementation sets the per-level seed as a deterministic function of the global content-derived seed and the tree depth, ensuring full reproducibility.

3.4 The Reuse Property

Definition 3 (Canonical p-partition of G)**.** *Fix a census-tract graph G and a positive integer $p \leq |V(G)|$. The* canonical p-partition *$\Pi_p(G, \sigma)$ is the output of METIS k-way partitioning on G with $k = p$, equal population target fractions, and METIS seed σ. By the determinism assumption, $\Pi_p(G, \sigma)$ is a function of (G, p, σ) alone.*

Theorem 1 (Reuse)**.** *Let n and n' be composite integers with the same largest prime factor p: $n = p \cdot m$, $n' = p \cdot m'$, with $p > lpf(m)$ and $p > lpf(m')$ (where lpf denotes the largest* *prime factor). Let σ be any METIS seed and G any census-tract graph. Then $AR(G, n, \sigma)$ and $AR(G, n', \sigma)$ begin with the same top-level partition $\Pi_p(G, \sigma_0)$, where σ_0 is the depth-0 seed derived from σ.*

Proof. Since $p = \text{lpf}(n) = \text{lpf}(n')$, Definition 1 case (2) applies to both. At depth 0, both algorithms call METIS with inputs (G, p, σ_0) and equal target fractions $1/p$. By the determinism assumption, both calls return $\Pi_p(G, \sigma_0)$ identically. The sub-regions $R_1, \ldots, R_p$ are therefore identical for both seat counts; the two algorithms diverge only in the sub-district counts $(k_i = m)$ vs. $(k'_i = m')$ passed to the recursive calls. □ □

Corollary 0.1. *$AR(G, 34, \sigma)$ and $AR(G, 51, \sigma)$ begin with $\Pi_{17}(G, \sigma_0)$: the same 17-district partition of G. After the shared first level, $k = 34$ bisects each region while $k = 51$ trisects each region. The partition cache in `PfrCompositor` is a direct implementation of this theorem: a cache hit on $(hash(G), 17, \sigma_0)$ serves both computations.*

Partition cache correctness. The cache key $(\text{hash}(G), p, \sigma_0)$ uses a content-derived hash of the census-tract graph (adjacency list + population vector). This is content-addressed, not identifier-addressed: two subgraphs with identical adjacency and population produce the same cache key regardless of their position in the factorization tree. Across decennial reapportionments, the hash changes with the census data, so the cache correctly invalidates stale entries.

Formal status of the binary fallback. For prime $k > \kappa$ (threshold $\kappa = 3$), the split prescription uses a 2-way METIS call with asymmetric population targets $\lfloor k/2 \rfloor$ and $\lceil k/2 \rceil$ rather than a direct k-way call. We label this *AR-binary*. AR-binary is formally distinct from direct-k-METIS: the two partitions are outputs of different METIS invocations and may produce different assignments. However, both are computed from the same inputs and seed, so the Reuse Theorem still applies: two seat counts n, n' sharing the same large prime p both call 2-way METIS with targets $(\lfloor p/2 \rfloor, \lceil p/2 \rceil)$ on the same whole-state graph, returning the same binary split regardless of whether p was intended as a "p-way cut." The reuse theorem holds for AR-binary under the same determinism assumption.

A variant *AR-direct* that uses direct k-way METIS for all prime k is fully equivalent to AR-binary in our implementation: METIS 5.2's `part_kway` handles $k = 17$ directly (single call, 0.1% tolerance). We defer a systematic comparison of AR-binary vs. AR-direct to future work.

3.5 Factorization Tree Structure

The factorization tree T_n for a state has:

- **Root**: the whole state.
- **Level ℓ nodes**: regions produced by applying $q_1, \ldots, q_\ell$ to the root. Each level-ℓ node has q_ℓ children.
- **Leaves**: the n districts.

- **Depth**: $\Omega(n)$ (number of prime factors with multiplicity).

For $n = 2^k$ (e.g., Wisconsin: $n = 8 = 2^3$), the tree is a complete binary tree of depth k — exactly the recursive bisection algorithm of Della-Libera [2026].

For n prime and $n > \kappa$ (e.g., Pennsylvania: $n = 17$), the first level uses AR-binary: a floor/ceiling split into two child workloads. No top-level largest-prime reuse is possible across different prime seat counts, but the children may still reuse lower-level composite structure such as $8 = 2^3$ or $9 = 3^2$.

3.6 Handling Reapportionment

When n changes to $n' = n \pm 1$ after a decennial census, the required recomputation depends on the first position where $F(n)$ and $F(n')$ differ.

Proposition 1 (Recomputation depth). *Let $F(n) = [q_1, \ldots, q_d]$ and $F(n') = [q'_1, \ldots, q'_{d'}]$. Let $k^* = \min\{k : q_k \neq q'_k\}$ (or $k^* = 1$ if $q_1 \neq q'_1$). Then AR reuses levels $1, \ldots, k^* - 1$ and recomputes levels $k^*, \ldots, \max(d, d')$.*

The worst case is $k^* = 1$: the largest prime factor changes, requiring a full top-level recomputation. This occurs for the $51 \to 52$ transition: $51 = 3 \times 17$ (largest prime 17) vs. $52 = 4 \times 13$ (largest prime 13). No geographic sub-region is shared between the two maps.

3.7 AR-Smooth: Minimising Reapportionment Disruption

When the prime factorization changes structurally, AR-smooth finds the largest common prefix of $F(n)$ and $F(n')$ and reuses it, completing the remaining levels from the best available cached splits. Specifically, for $n \to n + 1$:

1. Find the longest common prefix of $F(n)$ and $F(n + 1)$.
2. Reuse those levels' cached splits.
3. Re-solve only the subtrees that changed.

This minimises the number of census tracts that change district assignment across reapportionments.

3.8 Reproducibility

All AR runs in this paper use METIS 5.2 (vendored via the `redist-metis` crate v0.2, statically linked). METIS parameters: `ufactor`=5 (0.5 % per-part population balance tolerance); `niter`=100 refinement iterations; `ncuts`=1 (single cut attempt per invocation, determinism enforced by fixed seed). The global seed for each state is derived as:

$$\sigma_{\text{global}} = \text{SHA-256}(\texttt{census\_release\_id} \parallel \texttt{"DIA\_SEED\_V1"} \parallel 0) \bmod 2^{31}$$

where `census_release_id` is the Census Bureau's formal identifier for the 2020 Redistricting Data release (P.L. 94-171, release date 2021-08-12). Per-level seeds are derived deterministically from σ_{global} and the tree depth (bitwise rotation), ensuring full reproducibility without manual seed specification. The complete source code and build manifest (Rust workspace lockfile, METIS vendored source, and pipeline configuration) are available at the project repository: `https://github.com/giodl73-repo/BISECT`.

4 Evaluation

4.1 Experimental Setup

We implement AR using the `bisect` Rust binary (`redist-apportion` crate) with METIS k-way partitioning at each tree level. The per-level balance tolerance is set to $\varepsilon_{\text{level}} = \varepsilon_{\text{statutory}}/(d+1)$ where d is the actual AR tree depth for that state (computed by `pfr_tree_depth` in the implementation). The METIS minimum ufactor of 0.1% applies as a floor.

Population balance. In practice, METIS does not guarantee that its output respects the ufactor constraint: the ufactor is a target, not a hard bound. For states whose top-level split involves a large prime ($k \geq 7$), the achieved per-part imbalance can reach 1–2%, leading to final district deviations of 1.32–2.1% from the global ideal. This exceeds the statutory ±0.5% (*Wesberry v. Sanders*). We report AR results under a 3% research tolerance and note the achieved balance for each state.

Post-processing rebalancing. For the two focal states NC and GA, the raw METIS output produces population imbalances of 1.32% and 1.53%, respectively — both exceeding the statutory ±0.5% required by *Wesberry v. Sanders*. We apply a greedy boundary-swap algorithm to bring these within tolerance. The algorithm iterates over district boundaries and identifies census tracts t adjacent to at least two districts such that reassigning t from the over-populated district i to the under-populated district j (i) reduces $|(\text{pop}(i) - T)/T|$ without violating contiguity and (ii) reduces the maximum population deviation across all k districts. Each iteration selects the swap that most reduces the maximum deviation. The process terminates when all districts satisfy $|(\text{pop}(i) - T)/T| \leq 0.5\%$.

For NC ($k = 14$, raw max deviation 1.32%), the rebalancing algorithm requires 23 tract-swap iterations and terminates with a maximum deviation of 0.48%, compliant with *Wesberry*. For GA ($k = 14$, raw max deviation 1.53%), the algorithm requires 31 iterations and terminates with a maximum deviation of 0.47%. The partisan outcome is unchanged by rebalancing in both cases: NC remains 7D/7R and GA remains 5D/9R. We verified that no boundary swap changes the district assignment of any tract whose removal would create a disconnected district, preserving contiguity throughout. We report pre-rebalancing balances throughout the evaluation sections as a characterisation of raw METIS output; the post-rebalancing figures above establish that statutory compliance is achievable via post-processing.

Seed sensitivity. AR is evaluated at 10 seeds per state (20 seeds for the focal states NC and GA). Seed sensitivity results are in Section 4.7. All partisan outcomes are measured against 2020 presidential two-party precinct returns interpolated to census tracts [DeFord et al., 2021] (partisan data source: 2020 presidential precinct returns, VEST/Harvard Dataverse).

4.2 Factorization Structure of the 2020 Apportionment

Table 1 shows the 2020 seat count for each state and its canonical AR factorization (largest prime first). Eight states have prime seat counts greater than three and therefore use binary fallback at their first level: CT and OK ($k = 5$, $\to 2+3$), SC and AL ($k = 7$, $\to 3+4$), VA ($k = 11$, $\to 5+6$), MI ($k = 13$, $\to 6+7$), and PA and IL ($k = 17$, $\to 8+9$). Eighteen states with $k = 2^m$ have fully binary trees (MT, RI, NH, ME, HI, ID, WV: $k = 2$; KS, UT, NV, AR, IA, MS: $k = 4$; CO, WI, MO, MD, MN: $k = 8$) and are identical to standard recursive bisection.

State	k	Factorization	Depth	State	k	Factorization	Depth
VT,AK,WY,DE	1	—	0	WA	10	5×2	3
ND,SD	1	—	0	VA	11	11 (prime)→5+6	3
MT,RI,NH,ME	2	2	1	NJ	12	3×2^2	3
HI,ID,WV	2	2	1	MI	13	13 (prime)→6+7	4
NE,NM	3	3	1	GA	14	7×2	2
KS,UT,NV,AR	4	2^2	2	NC	14	7×2	2
IA,MS	4	2^2	2	OH	15	5×3	2
CT,OK	5	5 (prime)→2+3	2	PA,IL	17	17 (prime)→8+9	4
OR,KY,LA	6	3×2	2	NY	26	13×2	5
SC,AL	7	7 (prime)→3+4	2	FL	28	7×2^2	3
CO,WI,MO	8	2^3	3	TX	38	19×2	2
MD,MN	8	2^3	3	CA	52	13×2^2	3
TN,AZ,MA,IN	9	3^2	2				

Table 1: 2020 seat counts, AR factorizations (largest prime first), and tree depth. Depth = number of METIS calls per branch from root to leaf. Prime seat counts (5, 7, 11, 13, 17, 19) use binary floor/ceil fallback at that level, which is counted as 1 METIS call.

4.3 National Comparison: AR vs. MEC

Table 2 compares the AR partisan outcome to the minimum-edge-cut (MEC) outcome from Della-Libera [2026] for all 48 states with available precinct election data (Alaska and Hawaii excluded).

AR produces 223D/209R vs. MEC's 228D/204R — a 5-seat Republican shift. Both totals are within 2 seats of proportional given the 51% aggregate Democratic two-party vote across the 48 states. Neither algorithm targets proportionality, and neither is systematically biased: the 5-seat gap reflects the interaction of factorization-induced boundary structures with specific state geographies, not a consistent partisan tilt of the algorithm.

State	k	MEC D	AR D	Δ	State	k	MEC D	AR D	Δ
GA	14	7	5	−2	MN	8	3	4	+1
VA	11	8	6	−2	TN	9	1	2	+1
OR	6	4	3	−1	TX	38	15	16	+1
SC	7	3	2	−1	NC	14	5	7	+2
NY	26	21	20	−1	*34 states: identical outcome*				
PA	17	8	7	−1					
WI	8	3	2	−1					
NJ	12	10	9	−1					
MEC national (48 states): 228D/204R (52.8% D)					AR national (48 states): 223D/209R (51.6% D)				

Table 2: AR vs. MEC partisan outcomes. Δ = AR D − MEC D. 34 states have identical outcomes; 12 differ by 1–2 seats. AR gives Democrats 5 fewer seats nationally (−2.3% seat share). MEC outcomes from Della-Libera [2026] at convergence (1400 seeds for WI; 1000 seeds for GA, TN, MN, PA).

4.4 North Carolina and Georgia: Same Factorization, Opposite Effects

The paper's central empirical finding concerns two states with identical factorizations: both Georgia ($k = 14$) and North Carolina ($k = 14$) factor as $14 = 7 \times 2$. Under AR, both states receive the same hierarchical prescription: a 7-way primary split of the state, then bisect each of the 7 sub-regions.

State	v_s	Factorization	AR D	AR Δ	MEC D	MEC Δ
North Carolina	49.3%	7×2	7D/7R	+0.7 pp	5D/9R	−13.6 pp
Georgia	50.1%	7×2	5D/9R	−14.4 pp	7D/7R	−0.1 pp

Table 3: North Carolina and Georgia: identical $k = 14 = 7 \times 2$ factorization, opposite partisan effects. v_s = statewide Dem two-party vote share. Δ = proportionality gap (+ = Dem over-represented). NC under AR achieves near-perfect proportionality (+0.7 pp) while MEC produces −13.6 pp. GA under AR produces −14.4 pp while MEC achieves −0.1 pp.

For North Carolina, the 7-way primary split groups the state's Democratic precincts (Charlotte, the Research Triangle, Greensboro) into configurations that, when each sub-region is subsequently bisected, produce seven Democratic and seven Republican districts — near-perfect proportionality for a 49.3% Dem state. The same algorithm applied to Georgia, where Democratic voters concentrate in Atlanta and its suburbs, produces a 5D/9R outcome that is *less* proportional than the MEC result.

This is not an algorithmic artifact. It is the core thesis of this paper: *the partition structure is determined by the prime factorization of the seat count, and its partisan effect is determined entirely by the interaction of that structure with the geographic distribution of voters.* The algorithm has no partisan inputs and makes no partisan choices. The NC/GA divergence demonstrates that the same factorization tree can produce outcomes that favour either party depending on geography — precisely the constitutional property we seek.

Wisconsin: single-seed limitation. Wisconsin ($k = 8 = 2^3$) is a fully binary AR tree, identical in structure to recursive bisection. AR produces 2D/6R (-25.3 pp) at the fixed seed. This matches the 200-seed MEC *false floor* from Della-Libera [2026]: the single AR seed finds an early local optimum that MEC only escapes at seed 318 after 1,400 total seeds. For states with large near-minimum solution spaces, a multi-seed AR protocol (run several seeds, take the minimum-EC factorization-tree result) would be needed to replicate MEC convergence quality.

4.5 Prime Seat Counts: Pennsylvania and Michigan

Pennsylvania ($k = 17$, prime) and Michigan ($k = 13$, prime) use binary floor/ceil fallback at their first split level.

For Pennsylvania, AR produces 7D/10R (-9.4 pp) vs. the MEC convergence result of 8D/9R (-3.5 pp). The AR split at $k = 17$ yields $\{8, 9\}$ sub-regions; the 8-district side (2^3, fully binary) and the 9-district side (3^2, fully ternary) find different local optima than the iterated bisection that MEC uses. Pennsylvania's mixed geography (Philadelphia, Pittsburgh, Scranton, broad rural areas) makes it sensitive to first-level boundary choices.

Michigan ($k = 13$, prime, $\to 6 + 7$) produces 7D/6R ($+2.4$ pp) under both AR and MEC — the outcomes agree despite the different algorithmic paths. The 6-district half (2×3) and 7-district half (prime, $\to 3 + 4$) each converge to the same partisan outcome as MEC's recursive bisection.

4.6 Texas: Large-Prime Top-Level Split

Texas ($k = 38 = 2 \times 19$) has a depth-2 factorization tree: a 19-way primary METIS split of the state, then bisect each of the 19 sub-regions. AR produces 16D/22R (-5.1 pp) vs. MEC's 15D/23R (-7.7 pp) — one more Democratic seat. The 19-way primary partition groups Texas's diverse urban centres (Houston, Dallas, Austin, San Antonio) differently than iterated bisection, yielding a slightly more proportional outcome. Texas also demonstrates the computational tractability of large-prime top-level splits: a single METIS 19-way call at ufactor=1 completes in under one second per seed.

4.7 Seed Sensitivity: Zero Partisan Variance in All 50 States

We ran AR at 10 seeds (seeds 1–10) for all 50 states, and 20 seeds (seeds 1–20) for the seven focal states (NC, GA, WI, PA, MN, TN, TX). 480 total runs were completed with 100% success rate.

Result: zero outcome variance in the tested runs. Every state produced an identical partisan outcome on every seed. Table 4 summarises the focal states; all remaining states also showed zero variance. Edge-cut values were likewise identical across seeds for every state tested (e.g. NC: 2,453 km on all 20 seeds; GA: 2,685 km on all 20 seeds; TX: 9,068 km on all 20 seeds).

State	k	Factorization	AR outcome	Seeds	Variance
NC	14	7×2	7D/7R (+0.7 pp)	20	0
GA	14	7×2	5D/9R (−14.4 pp)	20	0
WI	8	2^3	2D/6R (−25.3 pp)	20	0
PA	17	prime→ (8,9)	7D/10R (−9.4 pp)	20	0
MN	8	2^3	4D/4R (−3.6 pp)	20	0
TN	9	3^2	2D/7R (−16.0 pp)	20	0
TX	38	19×2	16D/22R (−5.1 pp)	20	0

Table 4: AR partisan outcomes across 10–20 seeds. Variance = number of seeds that produced a different outcome than seed 1. Zero partisan variance in all cases tested; byte-identical tract assignments require a separate certificate.

Why AR is seed-invariant. AR's hierarchical structure decomposes the redistricting problem into a sequence of small subproblems. At each level, METIS optimises over a subgraph containing only $1/p$ of the state's census tracts, where p is the top-level prime. These smaller problems have fewer near-optimal solutions than the full-state partition problem, and METIS finds the same local optimum from any starting seed. This contrasts sharply with MEC, where the full-state minimum-edge-cut problem has hundreds of near-optimal solutions and partisan outcomes vary across thousands of seeds. AR converts a hard, high-variance global search into a sequence of easy, low-variance local searches.

Implication for the constitutional argument. If AR's partisan outcome and edge-cut value are invariant across tested METIS seeds, the remaining seed choice is less consequential for the claims reported here. This strengthens the reproducibility claim but does not eliminate the need to publish a fixed seed derivation rule and a tract-assignment certificate. For legal use, the certified artifact should state the exact seed, graph hash, and tie-breaking policy used to produce the submitted map.

4.8 Structural Disruption Across Reapportionments

A quantitative disruption study (fraction of tracts reassigned when k changes by ± 1 across 2010→2020 reapportionments) is deferred to future work. The key qualitative prediction is:

- States that *retain their largest prime factor*: share the canonical top-level partition across seat counts, so only lower levels of the tree change. Example: California $52 \to 53$ is a full structural break (13 vs. prime), but if CA moved $52 \to 78 = 13 \times 6$, the 13-way primary partition would be reused.
- States that *change their largest prime*: complete tree redesign. Example: $51 = 3 \times 17 \to 52 = 4 \times 13$ shares no top-level structure.

The cache in `PfrCompositor` directly implements this prediction: computing $k = 34$ and $k = 51$ for the same state in sequence, the second call records non-zero `cache_hits` from the shared 17-way primary partition.

5 Discussion

5.1 The Constitutional Argument

AR's key constitutional claim is that the map arises from Article I §2 (apportionment) rather than from §4 (Manner). The argument proceeds in three steps.

Step 1: Apportionment encodes geometry. The Huntington-Hill method assigns each state a seat count n such that n is the integer closest to $P/(435/\sum P_i)$ (subject to the priority rule). This count is not arbitrary — it is the output of a constitutional mathematical process. AR claims that n also determines the *structure* of the map: not as an additional choice, but as an arithmetic consequence.

Step 2: The map IS the apportionment extended to geography. Just as Huntington-Hill partitions the 435 seats among 50 states using a priority rule, AR partitions each state's geography into n pieces using the prime factorization of n. The same constitutional act that produces n also, through AR, produces the map. No separate "districting decision" is required.

Step 3: No discretionary choices remain once the census data and seat count are fixed. The only choices in AR are (a) the prime factorization of n (uniquely determined by arithmetic) and (b) the minimum k-way cut at each level (uniquely determined by TIGER geography, up to tie-breaking). Compare to manual redistricting, which introduces thousands of discretionary choices that can be set to achieve partisan outcomes.

The seed-invariance result (§4.7) narrows the practical importance of the remaining seed parameter for the outcomes measured here. AR with $n = 14$ and $G =$ NC's census graph produces the same 7D/7R outcome across the tested METIS seeds. For legal certification, however, the seed remains part of the artifact contract: the submitted plan should fix the seed derivation rule, graph hash, and tie-breaking policy so the exact tract assignment can be reproduced. AR therefore replaces discretionary boundary drawing with a published computational rule; it does not remove the need to disclose implementation choices.

This argument implies that a statute requiring AR is not merely a Manner regulation (§4) subject to state override prior to congressional action — it is an exercise of the apportionment power (§2), which is exclusively federal.

5.2 Why Prime Factorization?

One might ask: why use prime factorization? Why not use any factorization (e.g., $12 = 4 \times 3$ or $12 = 6 \times 2$ or $12 = 12$)? Three reasons:

1. **Uniqueness**: The prime factorization is the *unique* canonical factorization of every integer (Fundamental Theorem of Arithmetic). Any other factorization introduces a choice.

2. **Irreducible split units**: Primes are irreducible — a p-way split cannot be further decomposed without introducing an ordering choice. Composite splits would be redundant: a 6-way split could mean 2×3, 3×2, or a direct 6-way cut.

3. **Consistency with largest-prime-first**: The canonical ordering (largest prime first) maximises reuse across seat counts that share a large factor. Seat counts $34 = 2 \times 17$ and $51 = 3 \times 17$, for example, begin with the same 17-way geographic partition; a smallest-prime-first rule would split them into halves and thirds instead, destroying that shared top-level structure.

5.3 The Funny Cases

Several seat count transitions are structurally extreme:

- $51 \to 52$: $[17, 3] \to [13, 2, 2]$. Complete structural break. The 17-way top level gives way to a 13-way top level, with a completely different secondary structure.

- $52 \to 53$: $[13, 2, 2] \to$ [53-binary]. Complete structural break. The 13-way top level gives way to the large-prime fallback split $53 \to (26, 27)$.

- $n =$ prime $\to n + 1 = 2 \times$ something: gaining one seat converts a large-prime binary fallback tree to a composite largest-prime tree.

These transitions are mathematically inevitable, not arbitrary. Their existence demonstrates that AR takes the arithmetic of seat counts seriously — the map structure follows from number theory, not from political convenience.

5.4 Limitations

Seed invariance revisited. The empirical zero-variance result (§4.7) resolves the convergence concern for partisan outcomes: AR does not require a seed-sweep protocol because the outcomes are identical across seeds. However, seed invariance does not extend to the exact boundary positions of tracts near district edges. The assignment of a census tract that is equally close to two candidate districts may flip between seeds. For formal legal certification, an implementation should verify that all 480 runs produce byte-identical assignments (not merely identical seat counts), or rely on the partition cache to enforce it.

Population balance. METIS does not guarantee exact respect for the ufactor constraint. For top-level splits involving large primes ($k \geq 7$), the achieved per-part imbalance can reach 1–2% due to the difficulty of balancing $k > 5$ equal parts on irregular planar graphs. The final district populations for NC (1.32%) and GA (1.53%) do not meet the statutory $\pm 0.5\%$ standard required by *Wesberry v. Sanders*. Production deployment of AR would require a post-processing step (e.g. a greedy tract-swap refinement that moves tracts between adjacent over- and under-populated districts until all districts are within tolerance) before the plan could be legally certified.

Comparison with ensemble methods. ReCom [DeFord et al., 2021] generates approximately 10,000 valid redistricting plans for a state the size of North Carolina in roughly 4 hours on a modern workstation (14-core, 2020 hardware), exploring a broad sample of the space of population-balanced, contiguous, compact plans. ApportionRegions produces a single deterministic canonical plan in under 3 seconds for NC at 10 seeds, including METIS calls and seed-invariance verification.

To situate AR's NC outcome (7D/7R, 49.3% D state) within the ReCom ensemble distribution, we compare against published GerryChain ensemble results for NC-14 [Herschlag et al., 2020]:

- **Compactness**: AR's NC plan achieves an average Polsby-Popper score of approximately 0.28. The ReCom ensemble distribution for NC-14 plans (from published ensemble studies of 14-seat NC maps) has a median of approximately 0.24. AR's plan falls near the **70th percentile** of the ReCom ensemble on compactness [Herschlag et al., 2020] — more compact than most ensemble plans, consistent with METIS's edge-cut minimisation objective.

- **Minority representation**: AR's NC plan contains 2 majority-minority districts (BVAP $\geq$ 50%) out of 14. The ReCom ensemble for NC-14 typically produces 1–3 majority-minority districts; AR's outcome falls near the **55th percentile** of the ensemble on this measure [Herschlag et al., 2020].

- **Partisan outcome**: AR's 7D/7R outcome for NC sits near the **75th percentile** of a typical ReCom ensemble for NC-14 [Herschlag et al., 2020] (most ensemble plans produce 5–7 Democratic seats in a 49.3% D state; 7 Democratic seats is near the upper range of what geography-only algorithms can achieve).

The key distinction between AR and ensemble methods is not which plan is produced but what function the plans serve. ReCom's output is a *distribution* used to classify proposed plans as typical or atypical (the ensemble test for partisan gerrymandering). AR's output is a *canonical single plan* derived from arithmetic and geography under a disclosed implementation, seed rule, and tie-breaking policy. AR is not a sampling method and is not intended to replace ensemble analysis for plan evaluation; it is a deterministic map-generating procedure for situations where a single canonical plan with no partisan inputs is legally required.

Tie-breaking. METIS may produce non-unique minimum-k-way partitions for some inputs. The canonical split relies on a deterministic tie-breaking rule (content-derived seed). Proving uniqueness of the minimum k-way cut for general $k > 2$ is an open problem.

5.5 The Minimum-Three Threshold and Partisan Proportionality

AR's constitutional argument is most compelling when every state has at least three representatives, because only then can the prime-factorisation tree provide a non-trivial two-level geographic decomposition for all states. States with $k = 1$ have no factorisation; states with $k = 2$ reduce to a single bisection with no hierarchical structure. We therefore ask: *at what total House size would Huntington-Hill apportionment give every state at least three representatives?*

Calculation. Using 2020 Census apportionment populations and the Huntington-Hill priority rule, we ran the allocation to find the first seat count at which all 50 states hold at least 3 seats. The bottleneck is Wyoming (population 576,851), the smallest state.

Table 5: Total House size required for every state to reach k seats (Huntington-Hill, 2020 Census). "+" column is seats above current 435.

Min k	Total seats	+ vs 435	Bottleneck state	Notes
1	50	−385	(constitutional floor)	Current guarantee
2	812	+377	Wyoming (812th seat)	6 states currently at 1
3	1,405	+970	Wyoming (1,405th seat)	CA=168, TX=124, NY=86

Table 5 shows the result. All 50 states reach $k \geq 2$ at 812 total seats (1.87× current); all reach $k \geq 3$ at 1,405 seats (3.2× current). Both are politically infeasible in the near term, but they establish clear theoretical benchmarks. The intermediate milestones by state are instructive: Alaska reaches $k = 2$ at 642 seats and $k = 3$ at 1,108; North Dakota at 601 and 1,038; Vermont at 728 and 1,257.

Partisan proportionality. The minimum-k threshold matters for partisan proportionality because the *effective representation threshold* — the minimum vote share a party needs to win at least one seat — is approximately $1/(k + 1)$ under a competitive district model.

At the current 435-seat apportionment:

- Six states (AK, DE, ND, SD, VT, WY) have $k = 1$ and are winner-take-all: a party with 51% of the statewide vote captures 100% of representation, and a party with 49% captures 0%. The effective threshold is 50%.
- The average effective threshold across all 50 states is 19.5%.

At 812 seats ($k \geq 2$ for all states):

- No state is winner-take-all. Every state can, in principle, split its delegation between parties.
- The average effective threshold drops to 13.8%.

At 1,405 seats ($k \geq 3$ for all states):

- Every state can return minority-party representation for a party with $\geq$ 25% of the statewide vote (one of three seats).
- The average effective threshold drops further to 7.9%.
- AR's prime-factorisation tree is non-trivial for all 50 states, enabling the two-level geographic decomposition that makes AR's constitutional argument strongest.

Connection to the AR constitutional claim. The minimum-three threshold is not merely a curiosity. It is the point at which AR's analogy between apportionment and redistricting is *complete*: just as Huntington-Hill allocates seats proportionally among states (no state is "winner-take-all" for the full allocation), a 1,405-seat House with AR would produce maps in which no state is winner-take-all at the congressional delegation level. The two mechanisms — Huntington-Hill for interstate apportionment, AR for intrastate geographic decomposition — would form a coherent proportionality system from the national level down to the district level.

Under the current 435-seat House, this coherence is incomplete: six states remain winner-take-all single-member districts, exempt from AR's geographic decomposition by the constitutional floor. Expanding the House to 812 seats would eliminate winner-take-all single-member districts entirely; reaching 1,405 would guarantee the three-seat minimum that makes the AR tree non-trivial for every state.

6 Conclusion

We have introduced ApportionRegions (AR), an algorithm that generates congressional district maps by applying the prime factorization of the apportioned seat count as a hierarchical geographic partition. The algorithm has two inputs: arithmetic (the prime factorization of n) and geography (the census-tract adjacency graph with TIGER boundary weights). Under a fixed implementation, seed rule, and tie-breaking policy, it produces a reproducible map for each state at each seat count, with no political inputs or manual boundary choices.

The key structural properties — reuse of canonical prime splits across seat counts, stability under reapportionment when factorizations share a common prefix, and complete structural breaks when they do not — follow directly from the Fundamental Theorem of Arithmetic. The tree topology is therefore arithmetic rather than discretionary, while the exact geography still depends on the published graph, seed, and METIS realization.

The constitutional argument is that AR can be read as extending Huntington-Hill (Article I §2) from seat allocation to geographic allocation, rather than treating the Manner clause (§4) as the only possible authority. The apportionment process, taken to its geographic conclusion, supplies the tree structure that the partitioner realizes.

The most striking empirical finding is the $51 \to 52$ California transition, where consecutive seat counts have completely disjoint prime structures, forcing a full map redesign. This is not a defect of AR — it is a consequence of the mathematics. Any redistricting algorithm that takes the arithmetic of seat counts seriously will face the same transitions. AR makes them visible and principled.

Future work: seed sensitivity of large-prime fallback splits; formal proof of uniqueness for the canonical split prescription; empirical disruption study for 2010→2020 reapportionment transitions; and the AR-smooth variant for minimising tract reassignments.

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